

multiple applications. Garadi adds, "There is significant work on AC-to-DC, DC-to-AC and DC-DC power converters. You see them in all sectors, including industrial, home and even automotive too."

Current scenario of power electronics for engineers in India

"For a few years, there was not much focus on this field as the need was not felt. But today with so many developments happening, the scenario is changing," says Nayak.

Sharing his personal experience, he adds, "In a multinational company that I worked for earlier, I was responsible for developing a resource base for both R&D and engineering services catering to global requirements. It was difficult in the beginning to get the resources but in a span of two years we managed to hire and develop over 200 resources, though there were only a few experts in this area."

In India, as there are very few institutes/universities that offer competitive and valuable undergraduate or graduate programmes in power electronics, there is a scarcity of expert manpower in this field, said Vivek Agarwal, professor, Department of Electrical Engineering, Indian Institute of Technology-Bombay.

Usually, power electronics com-

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panies with indigenous capabilities belong to the small-scale or medium-scale category and do not have significant R&D activity.

Agarwal adds, "During the last two decades, some of the multinational companies have set up their R&D centres and manufacturing units in India where specialised power electronics systems are designed and developed but the number of such organisations is very small."

What's it like for freshers?

There are quite a few companies engaged in power electronics products, especially those in energy sector.

Nayak says, "Entry-level roles could be in technical areas like design, testing and commissioning. Those with a higher proficiency level could move into the research area."

He adds, "Nanotechnology is also bringing in tremendous opportunity in power semiconductor design." There are also some activities in automotive OEMs and Tier-1 companies in the areas of electric and hybrid vehicles.

"Currently, due to the scarcity of good power electronics engineers, the need is met by recruiting electronics or electrical engineers. Specialised (postgraduate-level) power electronics engineers are usually absorbed by multinational companies in R&D jobs and get to work on specialised projects," adds Agarwal. He believes that it is not difficult for a fresher with power electronics background to get into this sector.

On a slightly contradictory note, Garadi says, "It is highly unlikely that undergraduate freshers will be able to directly involve in power electronics design activities. However, smart engineers with a flair for power electronics could get into it."

"With on-the-job training under

the guidance of experienced engineers, freshers would do well by starting off with product testing and gaining experience donning the role of a design engineer," Garadi adds. Postgraduates in power electronics can plunge into design or debugging of power electronics products.

Typical pay packages

Power electronics field offers highly satisfying pay packages. Since it is a specialised segment, competent engineers enjoy a special status vis-à-vis other hardware engineers.

"In fact, they command a better pay package than mixed-signal board designers. The exact pay package depends on the company recruiting, plus the specific skills that the company is seeking. The starting salary for a postgraduate in power electronics could be around Rs 600,000 a year," Garadi informs.

"For undergraduates passing out from reputed academic institutes and universities, it is approximately Rs 400,000-700,000 a year. For postgraduate students, it is typically Rs 500,000 to Rs 1,000,000, while those with a PhD are able to draw Rs 900,000-Rs 1,600,000 a year," adds Agarwal.

Skills required

According to industry experts, applied mathematics will go a long way for a good grasp of power electronics. A good background on linear and non-linear systems will help in understanding the various concepts. Circuit simulation will help in reinforcing theoretical concepts.

Garadi suggests, "A good understanding of the behaviour of magnetics, power FETs and IGBTs is the key to designing and debugging power circuits but, finally, a hands-on approach is the only way to master any topic."

Skills required to succeed

1. A thorough knowledge of power electronics fundamentals, including basic power conversion topologies
2. Modelling of power electronics circuits and systems
3. Matlab/Simulink (or equivalent software) expertise for simulating power electronics systems in various applications
4. Control implementation
5. Knowledge of PCB design
6. Awareness about EMI/EMC issues
7. Programming of microcontrollers, DSP processors, etc
8. Knowledge of the hardware design of power electronics circuits and systems

—Vivek Agarwal, professor,
Department of Electrical Engineering,
Indian Institute of Technology-Bombay